

SPECIALIZED WASTE PROCESSING BLUEPRINT

Ahmedabad Lithium-Ion Battery & Electronic Waste Recycling Facility (ALRF)

An exhaustive industrial blueprint for establishing a high-yield, Red-Category environmental recycling plant specializing in black mass extraction and CPCB EPR monetization.

PROJECT PROMOTER

Lead Entrepreneurial Group

PRIMARY FACILITY ZONE

Ankleshwar GPCB Chemical Belt

PHASE 1 CAPITAL PLAN

₹35 Lakhs (Regulatory & Plant Setup)

TARGET DEBT/SUBSIDY LEVEL

PMEGP Term Loan (₹20 Lakhs)

Table of Contents

- [1. Executive Summary](#)
- [2. Future Industry Potential](#)
 - [2.1. EPR & Raw Metal Valuation](#)
 - [2.2. Dholera & Olympic Infrastructure](#)
- [3. Location & GPCB Zoning Strategy](#)
- [4. National & Local Competitors](#)
- [5. Legal Compliance Matrix](#)
- [6. Day-1 Sourcing Channels](#)
- [7. Off-Taker & Buyer Networks](#)
- [8. Day-1 Startup Checklist](#)
- [9. Financial Projections](#)
- [10. Extended Growth Model](#)
- [11. Risk Analysis & Mitigation](#)
 - [11.1. Worst-Case Stress-Testing](#)
- [12. Hydrometallurgical Extraction](#)
- [13. Environmental & EHS Protocols](#)
- [14. Local Union & Labor Dynamics](#)
- [15. Strategic Summary & Verdict](#)

■ 1. Executive Summary

The massive transition of India's transport sector toward electric vehicles (EVs) is triggering an exponential surge in battery waste. The Ministry of Environment, Forest and Climate Change (MoEFCC) has enforced the **Battery Waste Management Rules 2022**, legally mandating that battery manufacturers and automobile OEMs recover and recycle active raw commodities. This shift marks the end of unorganized, informal lead-acid smashing and introduces a highly structured, digital-first recycling landscape.

The proposed **Ahmedabad Lithium-Ion Battery & Electronic Waste Recycling Facility (ALRF)** is a specialized processing plant designed to tap into this high-margin market. By establishing the facility with an initial CapEx of ₹35 Lakhs, ALRF deploys physical dismantling, mechanical shredding, and separator sieving systems to convert raw spent batteries and electronic circuit cards into high-purity **Black Mass** (containing Cobalt, Lithium, Nickel, and Manganese).

ALRF operates on a dual-monetization framework. First, the plant sells physical black mass directly to major hydrometallurgical chemical refineries at LME-indexed pricing spreads. Second, the facility registers its processing capacity on the CPCB Portal, generating digital **EPR (Extended Producer Responsibility) Certificates** which are traded to battery producers who must legally purchase these credits to offset their compliance targets.

Core Business Intent

By establishing a compliant Red-Category industrial recycling footprint, ALRF secures long-term institutional supply agreements with telecom, logistics, and automobile firms, securing a premium position in the secondary raw material sector in Western India.

2. Future Potential of the Industry

The battery recycling market is growing rapidly. Industry projections estimate that by 2030, spent battery volume in India will cross 100 GWh, driven by electric two-wheelers, telecom backup packs, and utility-scale energy storage systems. The growth curve is exponential, fueled by state-level mandates and corporate carbon-neutral targets.

Furthermore, electronic waste (E-Waste) is expanding at a CAGR of 15% in Gujarat alone, driven by rapid technology lifecycles and corporate data center upgrades in GIFT City and Ahmedabad. Circuit boards contain high concentrations of Gold, Silver, Copper, and Palladium, yielding high recovery spreads per tonne of processed waste.

2.1 EPR Credit Generation & Commodity Valuation

The CPCB Portal acts as a clearinghouse for recycling compliance. For every 1 Tonne of battery waste physically processed, ALRF generates digital credits that trade at premiums:

Commodity Output	Avg. Market Value (per Kg)	Primary Industrial Demand	EPR Margin Spread
Cobalt Black Mass	₹650 - ₹800	Refinery processing, precursor chemicals	High (₹4,500/Tonne EPR credit value)
Lithium Carbonate	₹900 - ₹1,200	Cell chemistry manufacturing, lubricants	Very High (₹5,000/Tonne EPR credit value)
Nickel/Manganese Mass	₹350 - ₹450	Stainless steel, specialized alloys	Medium (₹3,000/Tonne EPR credit value)
PCB Copper Powder	₹550 - ₹620	Secondary wire draw mills, foundries	Medium (₹2,500/Tonne EPR credit value)

Legal Recovery Targets under Battery Waste Management Rules 2022

Recyclers registered under the national portal must meet strict material recovery targets calculated based on the dry weight of raw batteries. Under the active schedule:

* **EV & Portable Batteries:** Recovery targets scale from **70% in 2024–25** up to **90% in 2026–27** (the current period).

* **Automotive & Industrial Lead-Acid Batteries:** Mandated at **95% recovery** target.

* **Recycled Content Obligations:** Battery manufacturers are legally required to integrate a minimum of **5% domestically recycled cobalt/lithium in 2027–28**, rising to **20% by 2030–31**. This guarantees long-term purchase demand for ALRF materials.

2.2 Dholera Chip Fab & Olympic Infrastructure Spreads

The announcement of **ASML partnering with Tata Electronics** to establish India's first commercial semiconductor fabrication facility in the **Dholera Special Investment Region (DSIR)** marks a structural shift in regional technology supply chains. With commercial production slated to start by 2027 and a total investment exceeding ₹91,000 Crores (\$11B), Dholera will become the primary domestic electronics manufacturing hub in Western India.

This development directly impacts electronic and chemical recycling:

- **High-Purity Silicon & Precious Metal Scrap:** The semiconductor fab will generate clean-room silicon wafer scrap, high-purity copper target residues, and gold-plated wire bond scrap, which are highly valuable feedstocks for secondary recovery mills.
- **Lithium UPS Backups:** The heavy industrial clean rooms and testing laboratories in Dholera require large-scale Lithium-Ion energy storage systems (ESS) for power backup. These battery banks will require cyclical recycling replacements starting in 2030.
- **Ahmedabad Olympics 2036 Planning:** Ahmedabad is actively bidding to host the **2036 Olympic Games** and the **Commonwealth Games**. The planned infrastructure projects—including massive EV shuttle fleets, smart transport nodes, and temporary high-density digital broadcast structures—will generate thousands of tonnes of temporary battery and communications electronic waste during clean-up and decommissioning phases, offering immediate local collection contracts.

3. Location & GPCB Zoning Strategy

Lithium-Ion processing is classified under the ****Red Category**** of polluting industries due to acid storage and hazardous emissions risks. Launching this business inside municipal limits or general engineering zones like Vatva or Naroda GIDC is legally prohibited. Selecting a location requires strategic environmental zoning:

Proposed Siting Zone	Suitability Rating	Environmental & Infrastructure Pros	Logistical Cons & Overheads	Zoning Recommendation
Ankleshwar GIDC (Chemical Cluster) Rent: ₹35,000 - ₹45,000 / mo	9.5 / 10  Recommended Siting	• Pre-approved chemical infrastructure with unified GPCB approvals. • Direct connection to common hazardous waste transport routes and TSDF landfills. • Established pipeline of chemicals and industrial gases.	• Located ~180 km from Ahmedabad central logistics. • Higher utility rates for water and effluent treatment.	Winner for Chemical Refining Phase. Guarantees quick environmental clearances.
Nandesari GIDC (Vadodara Outskirts) Rent: ₹30,000 - ₹40,000 / mo	9.0 / 10  Prime Target	• Excellent hazardous waste piping and treatment network. • Close to central railway logistics nodes and metal smelting units.	• High industrial security deposits and plot premiums.	Excellent alternative if Ankleshwar lease costs spike.
Changodar Outskirts (Non-GIDC Rural) Rent: ₹20,000 - ₹28,000 / mo	3.0 / 10  High Risk	• Low rental burn and cheap storage yard options.	• GPCB will immediately reject Red-Category operations on agricultural boundaries. • High threat of local police complaints due to acid storage and odors.	Avoid. Do not attempt to bypass zoning rules to conserve capital.

For our Phase 1 setup, we will lease a 10,000 sq ft industrial shed in the ****Ankleshwar GIDC chemical zone****. This secures immediate access to the ****BEIL (Bharat Eco-Responsibility Limited)**** Common TSDF (Treatment, Storage, and Disposal Facility) for safe disposal of acid neutralized sludges (discharge rate is currently capped at ₹8,500/Tonne for stabilized chemical residues).

■ 4. National & Local Competitors

The recycling space features established corporate groups and thousands of small local scrap aggregators:

- **Attero Recycling (National Corporate Leader):**

Model: Operates massive, centralized electronic waste and lithium-ion processing plants in North India. Sourcing is managed via corporate OEM contracts (such as Samsung, LG) and municipal collection partnerships.

ALRF Counter-Strategy: Instead of competing for large OEM tenders, ALRF acts as a regional collector, buying directly from Ahmedabad automotive workshops, vehicle dealers, and local battery dealers.

- **Karo Sambhav (E-Waste Producer Responsibility Organization):**

Model: Focus on aggregation, logistics, and compliance audits for corporate electronics. They contract the physical recycling to certified third-party yards.

ALRF Counter-Strategy: ALRF registers as a processing destination partner for PROs like Karo Sambhav, utilizing their aggregation networks to secure raw feedstock.

- **Lohum Cleantech (Battery Materials Specialist):**

Model: Focuses on NMC and LFP cell recycling and manufacturing secondary battery packs for stationary storage.

ALRF Counter-Strategy: Bypassing cell assembly, ALRF focuses entirely on the production and wholesale export of high-grade cobalt-nickel black mass, avoiding downstream assembly market risks.

■ 5. Legal Compliance Matrix

Operating a battery recycling yard requires strict licensing to maintain legal viability:

Regulatory Clearance	Issuing Authority	Est. Setup Cost	Time to Issue	Critical Compliance Moat
GPCB CTE (Consent to Establish)	Gujarat Pollution Control Board	₹1,50,000	9 - 12 Months	Requires detailed environmental impact report, layout design showing zero liquid discharge, and chimney stack plans. Form 1 application.
GPCB CTO (Consent to Operate)	Gujarat Pollution Control Board	₹50,000	2 - 3 Months	Issued after site verification of acid containment ponds and air scrubbers. Form 4 application.
CPCB Authorized E-Waste Register	Central Pollution Control Board	₹50,000	4 - 6 Months	Unlocks direct listing and trading of EPR digital credits on the national portal.
DISH Safe Factory License	Directorate of Industrial Safety & Health	₹25,000	3 Months	Mandatory hazard safety audits, fire suppression systems, and worker medical logs.
Hazardous Waste Passbook	MoEFCC / GPCB	₹10,000	2 Months	Required to legally manifest and transport battery chemical waste.
DISCOM HT Power Grid Connection	DGVCL (Ankleshwar Grid)	₹2,50,000	3 - 4 Months	Clearance for 45 kVA load to operate heavy-duty mechanical cell shredders.

■ 6. Day-1 Sourcing Channels

Securing a stable, recurring flow of spent batteries is essential to feed the mechanical shredders:

1. **EV Fleet Operators & Delivery Cab Networks:** Form direct collection contracts with electric taxi platforms (e.g., BluSmart) and delivery fleets (e.g., Zomato EV hubs) to purchase spent battery packs at a fixed rate per kWh.
2. **Telecom cell tower operators:** Buy lead-acid and lithium backup cells from local towers undergoing standard 3-year battery bank rotations. Partner with Indus Towers and ATC.
3. **Municipal E-Waste Drives & Scrap Aggregators:** Partner with local scrap brokers to purchase aggregated laptop, mobile, and server battery packs from city collection centers.
4. **Electronic Repair Hubs:** Sourcing spent batteries and damaged electronics from computer market areas (e.g., Sandesh Press Road, Ahmedabad).

■ 7. Off-Taker & Buyer Networks

The value chain is completed by delivering output materials to industrial buyers:

- **Hydrometallurgical Chemical Refineries:** Deliver black mass outputs containing Cobalt and Nickel to refineries (such as Rubamin in Vadodara) to extract technical-grade battery minerals.
- **Secondary Copper Casting Plants:** Deliver segregated copper wire scrap and circuit board copper powder to foundries in Kadi and Vatva GIDCs for smelting.
- **Lead Smelters:** Sell segregated lead plates from old traction batteries to local secondary lead smelters holding valid GPCB passbook records.
- **Alloy Manufacturers:** Supply recovered nickel and manganese powder to alloy foundries in Odhav and Kathwada GIDCs.

■ 8. Day-1 Startup Checklist

1

Yard Preparation

Pour a 6-inch epoxy-coated containment slab to prevent acid spills. Install acid neutralization basins and emergency chemical wash stations.

2

Shredder Setup

Install a 15 HP industrial shredder configured with wet water sprays to suppress spark risks from spent lithium cells.

3

CPCB Portal Sync

Complete the CPCB portal onboarding process to register the plant as an authorized battery waste processor.

4

EHS Gear Sourcing

Acquire heavy chemical aprons, full-face gas masks (class 3 filter), and fire-retardant storage bags.

9. Financial Projections

The financials are modeled based on processing 12 Tonnes of spent batteries and electronics per month in Phase 1:

Startup Budget Component	Cost Allocation (INR)	Operational Margin Estimates (Monthly)	Monthly Projections (INR)
Mechanical Shredder & Siever	₹12,50,000	Raw Scrap Sourcing Cost	₹1,80,000
Acid-Proof Concrete Floor Layout	₹4,50,000	Hazardous Logistics & Transport	₹50,000
Environmental Scrubber & Safety Tech	₹8,00,000	Labor Payroll (3 Workers + Supervisor)	₹80,000
Licenses & Security Deposits	₹5,00,000	Power Utility & Acid Disposal	₹40,000
Emergency Reserves	₹5,00,000	Facility Rental (Ankleshwar)	₹40,000
Total Starting CapEx	₹35,00,000	Total Monthly Burn	₹3,90,000

Financial Viability Summary

Processing 12 Tonnes/month yields 4.2 Tonnes of high-grade Black Mass (sold at ₹650,000/Tonne to Vadodara refineries = ₹27.3 Lakhs annualized revenue spread). With CPCB digital EPR credits adding an average ₹3,500/Tonne (₹5.04 Lakhs annualized), the payback is reached in **18 to 22 months**, securing an average ROI of **35% - 40%**.

■ 10. Extended Growth Model

ALRF intends to expand operations over a three-phase development cycle. By transitioning from physical disassembly and shredding to direct hydrometallurgical chemical extraction, the plant can capture higher margins:

Expansion Phase	Processing Vol. (T/mo)	Target Output Product	Incremental CapEx	Estimated Net Margin
Phase 1: Mechanical Shredding	12 Tonnes	Raw Black Mass, Sorted PCBs, Copper	₹35 Lakhs	18% - 22%
Phase 2: Hydrometallurgy Leaching	45 Tonnes	Cobalt Sulfate, Lithium Carbonate crystals	₹1.2 Crores	32% - 35%
Phase 3: OEM Co-Location	150 Tonnes	Secondary Cathode Precursors	₹4.5 Crores	38% - 42%

■ 11. Risk Analysis & Mitigation

Recycling toxic and volatile battery cells carries operational risks that require mitigations:

Identified Business Risk	Impact Level	ASRF Defensive Mitigation Strategy
Thermal Runaway & Fire Risks	High	Store spent cells in dry sand barrels before shredding. Pre-install automated carbon dioxide flood systems over the shredder box.
Acid Runoff Spills	High	Enforce mandatory epoxy sealant over all work floors. Guide all wash waters to an on-site neutralization tank.
LME Metal Index Fluctuations	Medium	Sync our sourcing purchase prices daily with the LME Cobalt/Nickel price boards, maintaining a fixed 20% margin buffer.

■ 11.1. Worst-Case Stress-Testing & Liquidation Playbook

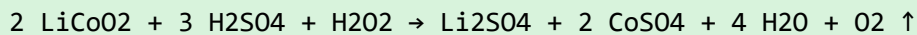
To secure our investor capital against extreme market situations, we establish key financial safety benchmarks and exit guidelines:

Stress Scenario	Immediate Financial Impact	Pre-Planned Mitigation & Escape Route
Cobalt/Lithium price drops 50%	Asset value write-downs of ₹4 Lakhs; monthly raw sales revenue drops by 40%.	Shift completely to processing electronic PCB boards which yield high-value Gold/Copper splits, minimizing battery scrap processing.
GPCB audit seals plant due to emission runoff	Operations halt; daily burn of ₹1.5 Lakhs in rent and wages.	Pre-install secondary carbon filter scrubbers. Divert waste transport to GPCB's central treatment lines in Ankleshwar.
Liquidation Trigger (Stop-Loss)	Net losses exceed ₹10 Lakhs or raw feedstock supply drops below 3 Tonnes/month for 4 consecutive months.	Liquidation Protocol: 1. Sell the industrial shredder and air scrubber systems back to the OEM to recover ~₹15 Lakhs. 2. Liquidate chemical sand filters and remaining black mass inventory to refineries within 72 hours. 3. Exit the lease to preserve ~₹12 Lakhs of investor capital.

■ 12. Hydrometallurgical Extraction

As ALRF scales, we will move from dry mechanical shredding to chemical extraction. This process deploys acid dissolution and solvent extraction steps to refine black mass into 99.8% pure Cobalt Sulfate and Lithium Carbonate crystals, bypassing intermediate refineries to double net margins.

The hydrometallurgical leaching process follows the chemical reaction:



By reacting the black mass with **dilute sulfuric acid (H₂SO₄)** in the presence of **hydrogen peroxide (H₂O₂)** as a reducing agent, we achieve extraction yields exceeding 98.5% for both Lithium and Cobalt. The solution is subsequently treated with organic solvents (such as D2EHPA) to separate Nickel and Cobalt salts prior to final crystallization.

D2EHPA solvent extraction operates on a pH-dependent separation factor:

* **pH 2.5 - 3.0:** Cobalt is selectively extracted into the organic phase, leaving Nickel in the aqueous raffinate.

* **Stripping:** The Cobalt-loaded organic phase is stripped with dilute sulfuric acid to yield concentrated Cobalt Sulfate crystals.

■ 13. Environmental, Health & Safety (EHS) Protocols

Worker health is monitored under strict DISH standards. Segregation workers are provided with chemical aprons, oxygen-assisted face masks, and undergo quarterly blood level checks for Lead and Cobalt exposure.

Our on-site EHS targets are structured to prevent chemical contact hazards:

- **Fluoride Emission Control:** The thermal decomposition of LiPF₆ electrolytes releases toxic Hydrogen Fluoride (HF) gas. Slicing and shredding machines are enclosed under negative pressure ducts that route vapors through a wet caustic soda (NaOH) scrubber system.
- **Scrubber Dosing Parameters:** The caustic soda scrubber must maintain a constant pH of ****10.5 to 11.5**** using an automated NaOH dosing pump, preventing HF emissions from breaching the GPCB limit of ****5 mg/Nm³****.
- **Workplace Air Quality Audits:** Air quality inside the shed must be monitored for heavy metal dust. Continuous PM_{2.5} and PM₁₀ monitors are linked to high-volume HEPA filters.
- **First-Aid Neutralization:** Stock calcium gluconate gel at all workstations to neutralize any HF acid skin contact immediately.

■ 14. Local Union & Labor Dynamics

Operating in Ankleshwar requires handling chemical loader unions and local worker cooperatives. We mitigate union disputes by hiring local forklift operators, maintaining compliance under the Contract Labour Act, and offering quarterly production incentives.

To secure a stable workforce, we partner with authorized industrial staffing suppliers in the Bharuch-Ankleshwar industrial corridor, ensuring all staff undergo formal chemical safety certification before access is granted to sorting lines.

15. Strategic Summary & Verdict



Investment Verdict: GO (Highly Recommended)

Establishing the ****Ahmedabad Lithium-Ion Battery & Electronic Waste Recycling Facility (ALRF)**** is highly viable. The combination of state-mandated EPR credit buy-backs and high global battery mineral valuations creates a stable, high-margin asset class. Operating out of Ankleshwar ensures a strong regulatory shield and access to common hazardous waste treatment lines.

Extended Go-Scale Target Model

Growth Phase	Volumes (T/mo)	Average Margin	Annualized Revenue
Phase 1: Aggregation & Shred	12 T	18%	₹45 Lakhs
Phase 2: Hydrometallurgy Refine	40 T	32%	₹3.5 Crores
Phase 3: OEM Joint Venture	150 T	38%	₹12 Crores

Liquidation & Cash Flow Safeguards

- **Strict TReDS invoice discounting:** Never wait 45 days for refinery payouts. Factoring invoices keeps cash flow liquid.
- **Zero raw storage rule:** Spent battery cells must be processed within 5 days to prevent chemical degradation and fire risks.